THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 7 & 8: **How do Organisms Reproduce?, Heredity** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The pollen tube reaches the ovule through the: 1

(a) style **(b)** petal **(c)** sepal **(d)** filament

2. Potato is propagated by: 1

(a) root **(b)** spores
(c) leaf **(d)** stem tuber (eyes)

3. Menstruation lasts about: 1

(a) 2 to 8 days **(b)** 15 days **(c)** one hour **(d)** 28 days

4. The number of chromosomes in a human zygote is: 1

(a) 46 **(b)** 23 **(c)** 22 **(d)** 92

5. The movement of pollen from one flower to another of the same species is: 1

(a) fertilisation **(b)** germination
(c) self-pollination **(d)** cross-pollination

6. The term 'gene' refers to a section of DNA that: 1

(a) makes energy **(b)** stores fat
(c) digests food **(d)** provides information for one protein

7. Number of different gametes produced by RrYy is: 1

(a) 1 **(b)** 4 **(c)** 8 **(d)** 2

8. A woman has a perfect pair of sex chromosomes called: 1

(a) XO **(b)** XX **(c)** YY **(d)** XY

9. The traits that are transmitted from parents to offspring are called: 1

(a) inherited traits **(b)** learnt traits **(c)** mutant traits **(d)** acquired traits

10. In the F2 of a dihybrid cross of RRYY × rryy, the number of round green plants out of 16 is: 1

(a) 6 **(b)** 9 **(c)** 3 **(d)** 1

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Name the parts of the pistil. 2

12. What is the role of the cotyledons in a seed? 2

13. What are the two kinds of sperm produced by a human male? 2

14. State the law of segregation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What are the disadvantages of surgical methods of contraception? 3

16. How did Mendel find that traits are inherited independently? 3

17. Explain the role of tissue culture in horticulture. 3

OR

What would happen if gametes had the same number of chromosomes as body cells? 3

18. Explain why recessive traits appear in the F2 generation. 3

OR

What is the significance of having two copies of each gene in the body? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Why is variation important for a species?
(ii) How does reproduction lead to variation?
(iii) Give one example of variation. 5

OR

(i) Describe the female reproductive system.
(ii) Explain the events of the monthly cycle when the egg is not fertilised. 5

20. (i) Distinguish between dominant and recessive genes.
(ii) Explain with the help of a cross how a recessive trait is masked in F1. 5

OR

(i) Why do variations arise in sexual reproduction?
(ii) Why is variation beneficial?
(iii) Explain with an example from a population. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND